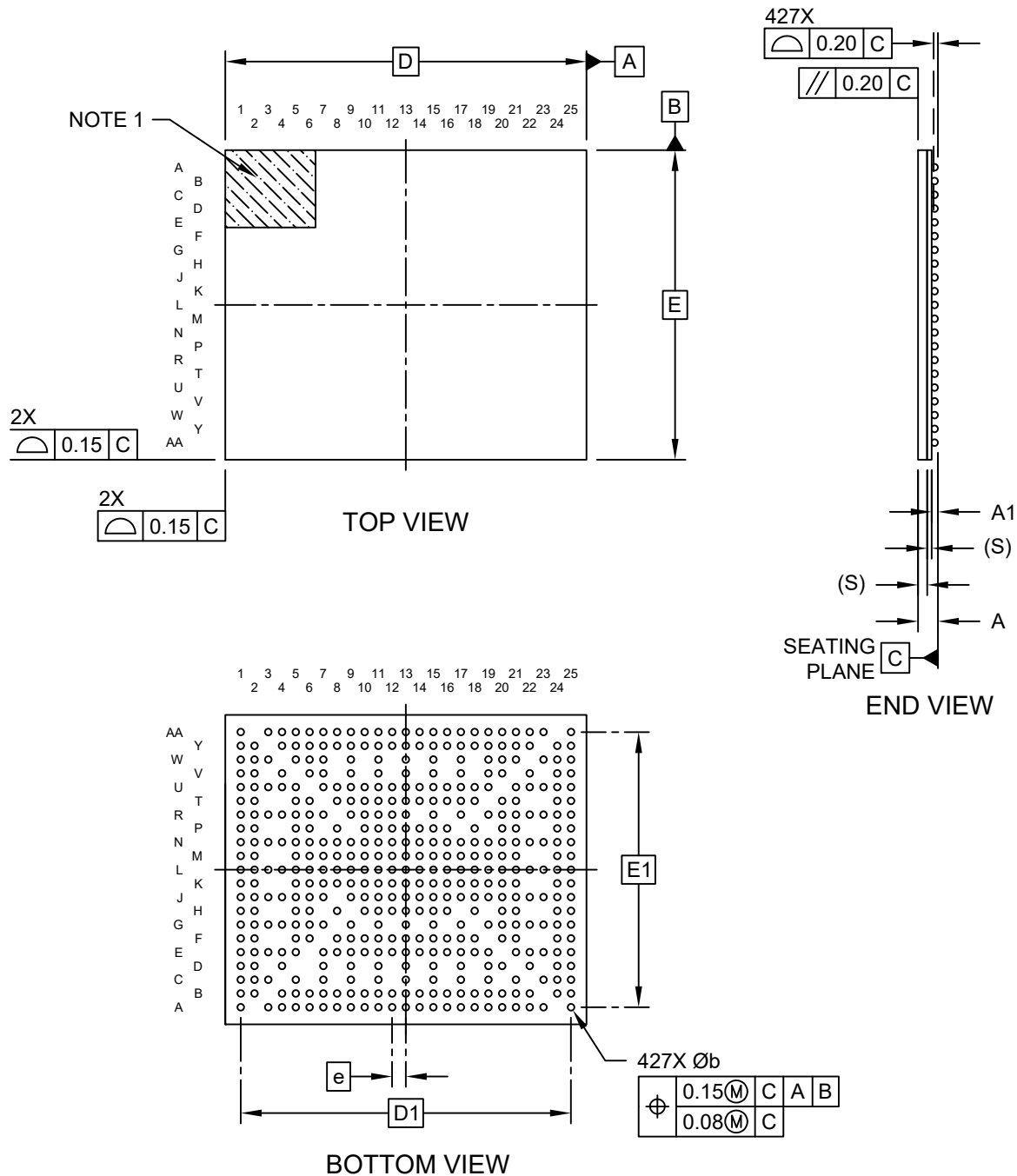


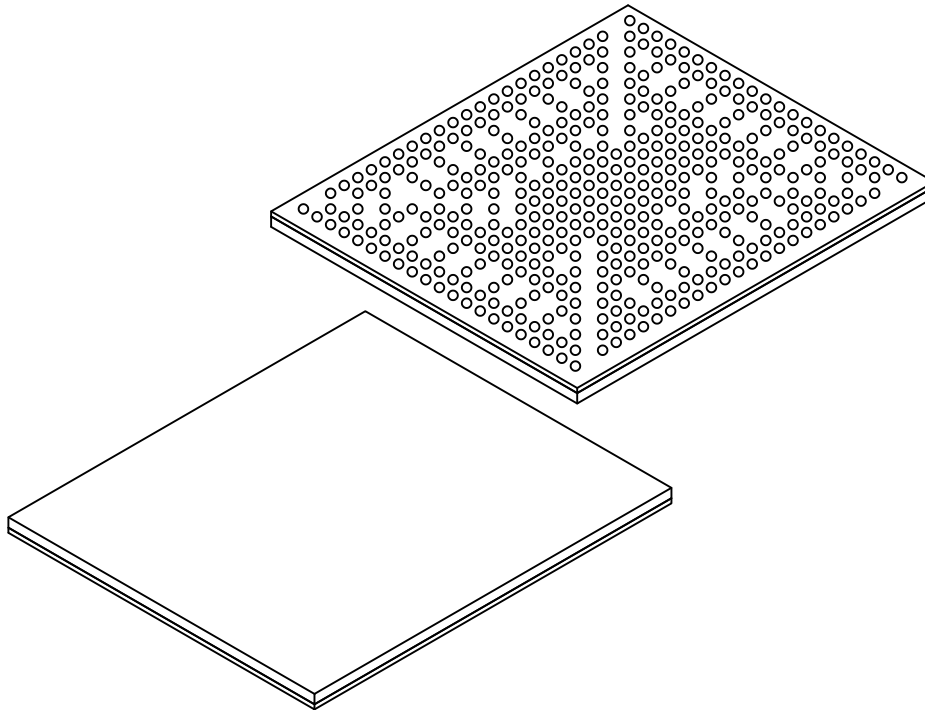
427-Ball Thin Fine-Pitch Ball Grid Array (4UB) - 21x18x1.2 mm Body [TFBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



427-Ball Thin Fine-Pitch Ball Grid Array (4UB) - 21x18x1.2 mm Body [TFBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



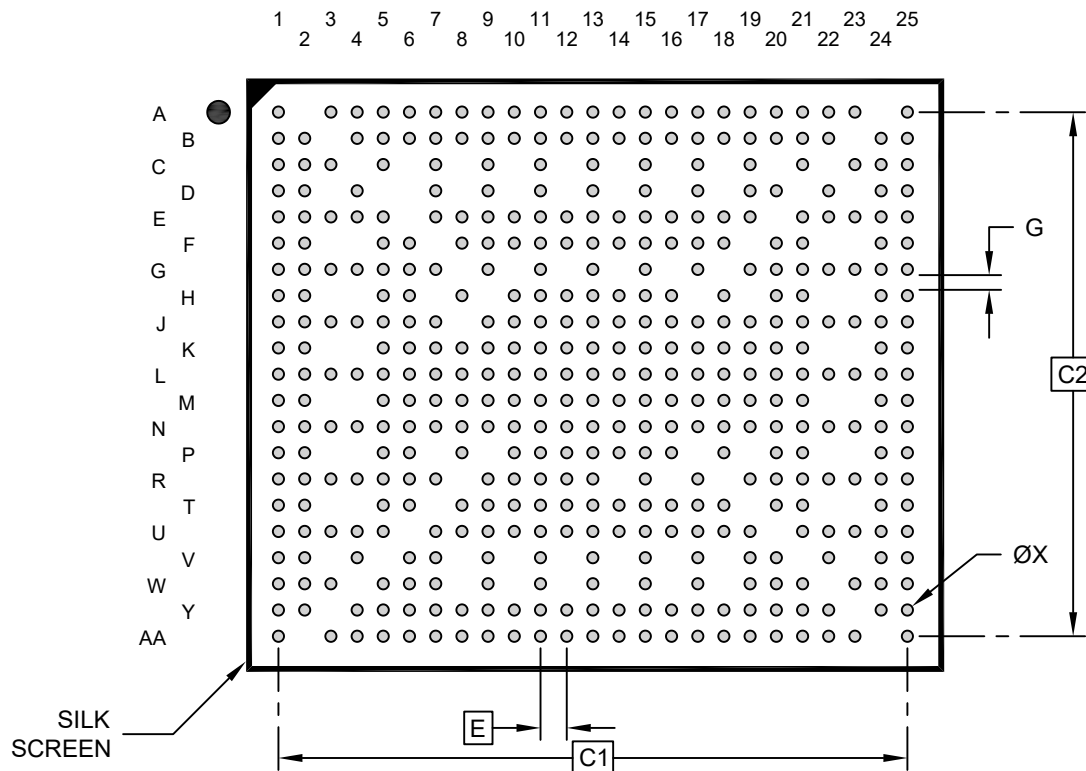
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	427		
Pitch	e	0.80 BSC		
Overall Height	A	–	–	1.20
Ball Height	A1	0.27	–	0.37
Mold Thickness	M	0.53 REF		
Substrate Thickness	S	0.26 REF		
Overall Length	D	21.00 BSC		
Ball Array Length	D2	19.20 BSC		
Overall Width	E	18.00 BSC		
Ball Array Width	E2	16.00 BSC		
Ball Width	b	0.38	–	0.45

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

427-Ball Thin Fine-Pitch Ball Grid Array (4UB) - 21x18x1.2 mm Body [TFBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.80 BSC		
Contact Pad Spacing	C1	19.20 BSC		
Contact Pad Spacing	C2	16.00 BSC		
Contact Pad Width (Xnn)	X			0.35
Contact Pad to Contact Pad (Xnn)	G	0.45		

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M

BSC: Basic Dimension. Theoretically exact value shown without tolerances.

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